

[illegible]

Let $S_n = 1^2 + 3 \times 2^2 + 3^2 + 3 \times 4^2 + 5^2 + 3 \times 6^2 + \dots + (2 + (-1)^n)n^2$.

(b) Show that $S_{2n} = \frac{1}{3}n(2n+1)(an+b)$, where a and b are integers to be determined. [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(c) State the value of $\lim_{n \rightarrow \infty} \frac{S_{2n}}{n^3}$. [1]

[illegible]